

Mike Mo

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Research Interests

My research investigates how to formally evaluate the quality of approximation algorithms. My recent work applied an information-theoretic metric to locality-sensitive hashing for sequence alignment. I am broadly interested in the complexity-theoretic limits of approximation and in theories of algorithms that reason under uncertainty and strategic interaction, particularly computational learning theory and algorithmic game theory.

Education

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Computer Science, GPA: 3.87/4.00

Aug 2025 – May 2029

Relevant Coursework: 15-251 (Great Ideas in Theoretical Computer Science), 21-425 (Probability and Martingales), 21-269 (Vector Analysis), 15-150 (Principles of Functional Programming)

Academic Honors: Honors math sequence (60 out of 800 freshmen), Dean's List (Fall 2025, Spring 2026)

Research Experience

SURA, Carnegie Mellon University

May 2026 – Jul 2026

Information-Theoretic Evaluation and Design of Locality-Sensitive Hashing for Protein Alignment

- Proposed an evaluation score based on Shannon conditional entropy for LSH algorithms that estimate sequence edit distance
- Implemented multiple LSH algorithms and a Rayon-parallelized benchmarking framework to evaluate their accuracy
- Proposed and implemented a novel LSH algorithm for protein alignment in the low-similarity regime, outperforming the strongest existing baseline by 1.7%

PROMYS, Boston University

Jun 2024 – Aug 2024

Generalizing Tiered Trees to Hypertrees

- Generalized tiered trees to hypergraphs, defining the resulting tiered hypertree structure
- Designed and implemented a C library to compute hypertree weights and to extend the Tutte polynomial to hypertrees
- Applied the extended Tutte polynomial to count hypertrees of a given weight under a fixed tier structure

Independent Research, ETiger

Jan 2022 – May 2022

Theoretical Analysis of the Accuracy of Suboptimal Path-finding Algorithms (ICACS 2024)

- Replicated an experiment to assess the accuracy of suboptimal path-finding algorithms for 2D grids with obstacles
 - Derived curves of algorithm accuracy with respect to obstacle densities using techniques from probability theory and dynamic programming
 - Proposed a methodology to select the best suboptimal algorithms based on obstacle density
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Publications

- [1] **Chenghao Mo.** “Theoretical Approach on Assessing the Accuracy of the Shortest Path Non-Optimal Algorithm for 2-Dimensional Grids with Obstacles”. In: *Proceedings of the 2024 8th International Conference on Algorithms, Computing and Systems*. ICACS '24. Association for Computing Machinery, 2025, pp. 150–156. ISBN: 9798400718304. DOI: [10.1145/3708597.3708620](https://doi.org/10.1145/3708597.3708620). URL: <https://doi.org/10.1145/3708597.3708620>.
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Teaching Experience

Teaching Assistant, Carnegie Mellon University Aug 2026 – Present
 21-127 (*Concepts of Mathematics*)

- Hold office hours to guide students through mathematical proof construction
- Grade and provide feedback on problem sets

Peer Tutor, Carnegie Mellon University Feb 2026 – Apr 2026
 15-122 (*Principles of Imperative Computation*)

- Provided targeted academic support for at-risk students
 - Developed personalized learning plans and practice problems to bridge gaps in programming concepts
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Technical Skills

- **Programming Languages:** Rust, Python, C, C++, Bash
 - **Frameworks & Libraries:** PyTorch, Pandas, Tokio
 - **Developer Tools:** Linux, Git, GitHub, GDB, Valgrind, CMake
 - **Scientific Computing:** MATLAB, GNU Octave, Wolfram Alpha, Maxima
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Selected Projects

News Summarizer [\[GitHub\]](#) Mar 2026
 Course Project

- Developed a concurrent data pipeline in Rust using tokio to scrape and normalize real-time headlines from multiple major news outlets
- Integrated a local LLM to automate news categorization, transforming raw text into structured data for rapid information synthesis

- Optimized the parsing engine to handle malformed HTML, reducing data loss and ensuring the integrity of the downstream NLP pipeline

NBA Betting Bot [\[GitHub\]](#)

Jan 2026

Personal Project

- Implemented a Python script using Pandas to fetch game data from the NBA API and managed storage via a SQL database
- Trained a Gradient Boosted Decision Tree (GBDT) using scikit-learn to predict results of future games, achieving an accuracy of 70%

Awards

Outstanding Achievement, 49th–51st NH State Mathematics Contests	2023, 2024, 2025
New Hampshire ARML State Team Qualifier	May 2024
High School's Only AIME Qualifier (among classes of 2021–2026)	Nov 2022
Bronze Medal, Asia-Pacific Informatics Olympiad (APIO)	Apr 2022
USACO Gold Division	Mar 2022
Bronze Medal, National Olympiad in Informatics Winter Camp (NOI WC)	Feb 2022
First Prize, Certified Software Professional-Senior (CSP-S)	Oct 2021

References

Guillaume Marçais, Senior Systems Scientist

Ray and Stephanie Lane Computational Biology Department, Carnegie Mellon University

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Research supervisor, SURA

Klaus Sutner, Teaching Professor

Computer Science Department, Carnegie Mellon University

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Instructor, 15-251

Noha Abdelghany, Assistant Teaching Professor

Department of Mathematical Sciences, Carnegie Mellon University

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Instructor, 21-127